

Internship Project Summary

End-to-End Sales Forecasting & Demand Intelligence System

Overview

Yeh project ek **real-world retail problem** solve karta hai jahan companies ko predict karna hota hai ki future me kitni sales hogi aur kitna stock maintain karna chahiye. Is system ka goal hai **data-driven decision making** enable karna — taaki business overstock aur understock dono problems se bach sake.

Is project me maine **time series analysis, machine learning, anomaly detection, clustering aur dashboard deployment** ko combine karke ek complete end-to-end solution build kiya.

Objective

- Future sales ka accurate forecast karna
- Unusual sales spikes/drops identify karna
- Products ko demand behavior ke basis pe segment karna
- Ek interactive dashboard banana jisse business team easily decisions le sake

Data Used

- **Primary Dataset:** Superstore Sales Dataset (4 years daily data)
- **Secondary Dataset:** Video Game Sales Dataset (multi-source analysis ke liye)

Key Work Done

1. Data Analysis & Exploration

- Data clean kiya (missing values, duplicates)
- Date features extract kiye (Month, Year, Season etc.)
- Sales ko daily se weekly & monthly level pe aggregate kiya
- Business insights nikale:
 - Top revenue category identify ki
 - Region-wise growth analyze ki
 - Seasonality patterns detect ki

2. Time Series Analysis

- Monthly sales trend visualize kiya
- Time series decomposition apply kiya:
 - Trend
 - Seasonality
 - Noise
- ADF test se stationarity check ki
- Differencing use karke data ko model-ready banaya

3. Sales Forecasting Models

3 different approaches implement kiye:

SARIMA (Statistical Model)

- Traditional time series forecasting
- Seasonal patterns capture kiye

Prophet (Industry Tool)

- Automatic trend & seasonality detection
- Business-friendly interpretation

XGBoost (ML Model)

- Lag features aur rolling averages use kiye
- High accuracy prediction

Models ko MAE, RMSE, MAPE se compare kiya
Best model select kiya based on performance

4. Segment-wise Forecasting

- Category level forecast (Furniture, Technology, Office Supplies)
- Region level forecast (West, East)
- Growth comparison kiya across segments

5. Anomaly Detection

- Isolation Forest use karke unusual sales detect ki
- Z-score method se validation ki
- Real-world reasons identify kiye:
 - Festive spikes
 - Sales campaigns

- Unexpected drops

6. Product Segmentation

- K-Means clustering apply kiya
- Products ko groups me divide kiya:
 - High demand stable
 - Low demand volatile
 - Growing products
 - Declining products

Har cluster ke liye **stocking strategy recommend** ki

7. Dashboard Deployment (Streamlit)

Ek interactive web app banaya jisme:

- Sales overview dashboard
- Forecast explorer
- Anomaly report
- Product segmentation insights

Business users easily filters use karke decisions le sakte hain

Key Results

- Accurate 3-month sales forecast generate kiya
- High-demand products identify kiye
- Seasonal sales patterns clearly detect hue
- Multiple anomalies successfully detect hue
- Business-ready dashboard successfully deploy kiya

Business Impact

Yeh system help karta hai:

- Inventory optimization
- Cost reduction (overstock avoid)
- Sales loss prevention (understock avoid)
- Better demand planning

- Faster decision making

Limitations

- Model historical data pe dependent hai
- Sudden market changes (COVID-type events) predict nahi kar sakta
- External factors (marketing, economy) include nahi kiye gaye

Conclusion

Yeh project ek **complete end-to-end data science solution** hai jo real-world retail problems solve karta hai. Isme multiple advanced techniques use karke ek practical system build kiya gaya hai jo directly business decisions me use ho sakta hai.